Research summary
NV/SiV-EIT project

June 25st 2025

The Budker group @JGU Mainz

Institute of SCIENCE TOKYO

Research question and Hypothesis

How can one increase the operating temperature of the diamond EIT using the ESLAC?

Sub issue 1: How does the lambda system of the ESLAC-induced NV-EIT differ from the strain-induced NV-EIT?

Sub issue 2: How robust is the ESLAC-induced EIT against perturbations of B-field and strain?

Sub issue 3: At which temperature does the ESLAC induced NV-EIT break down?

Agenda

- What is EIT and what has been done with NVs?
- Which excited states should be used for EIT?
- Is GSLAC useful for EIT?
- Relevance of the Faraday rotation paper to the room-temperature EIT

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Sub issue 1 How does the lambda system of the ESLAC-induced NV-EIT differ from the strain-induced NV-EIT?

Sub issue 1: How does the lambda system of the ESLAC-induced NV-EIT differ from the strain-induced NV-EIT?

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Last year

Defined excited-state Hamiltonian at low temperature

The excited states

Doherty et al Phys Rep (2013)

The excited-state bare Hamiltonian is

$$\hat{H}_{es}^{LT} = D_{es}^{\parallel}\left[\hat{S}_z^2 - \frac{S(S+1)}{3}\right] - \lambda_{es}^{\parallel}\hat{\sigma}_y \otimes \hat{S}_z$$
$$+ D_{es}^{\perp}\left[\hat{\sigma}_z \otimes (\hat{S}_y^2 - \hat{S}_x^2) - \hat{\sigma}_x \otimes (\hat{S}_y\hat{S}_x + \hat{S}_x\hat{S}_y)\right]$$
$$+ \lambda_{es}^{\perp}\left[\hat{\sigma}_z \otimes (\hat{S}_x\hat{S}_z + \hat{S}_z\hat{S}_x) - \hat{\sigma}_x \otimes (\hat{S}_y\hat{S}_z + \hat{S}_z\hat{S}_y)\right],$$

where $\vec{\sigma} = [\sigma_x, \sigma_y, \sigma_z]$ are the standard Pauli matrices that represent the fine structure orbital operators in the basis of orbital states $|^3E_x\rangle, |^3E_y\rangle$ associated with the ^3E level.

The interaction Hamiltonian is

$$\hat{V}_{es}^{LT} = d_{es}^{\parallel}(E_z + \delta_z) + d_{es}^{\perp}(E_x + \delta_x)\hat{\sigma}_z - d_{es}^{\perp}(E_y + \delta_y)\hat{\sigma}_x + \mu_B(l_{es}^{\parallel}\hat{\sigma}_y + g_{es}^{\parallel}\hat{S}_z +)B_z$$
$$+ \mu_B g_{es}^{\perp}\left(\hat{S}_x B_x + \hat{S}_y B_y\right),$$

where $l_{es}^{\parallel}$ is the ^3E orbital magnetic moment, $g_{es}^{\parallel}$ and $g_{es}^{\perp}$ are the components of the ^3E electronic g-factor tensor, and $d_{es}^{\parallel}$ and $d_{es}^{\perp}$ are the components of the 3E electronic electric dipole moment.

Sub issue 1: How does the lambda system of the ESLAC-induced NV-EIT differ from the strain-induced NV-EIT?

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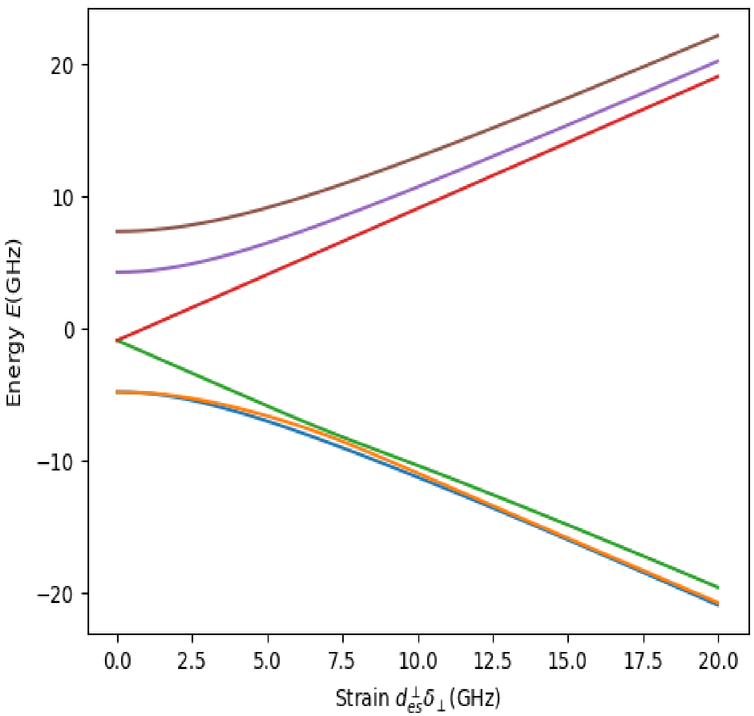
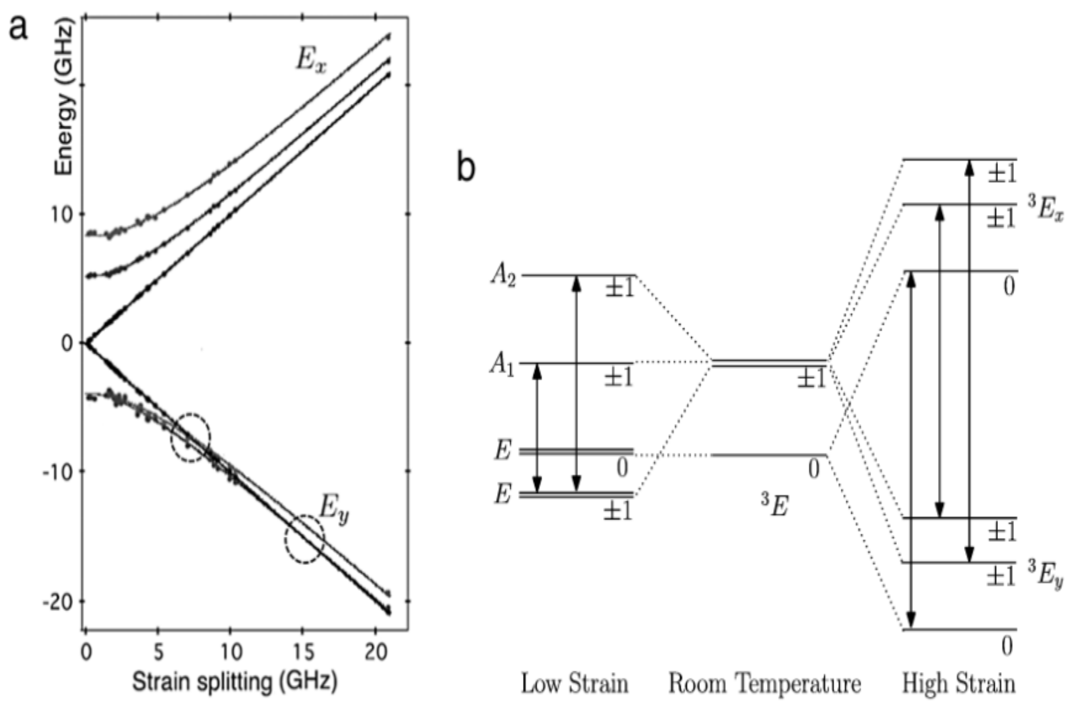
Last year

Defined excited-state Hamiltonian at low temperature

Our program



Doherty et al Phys Rep (2013)



Our program reproduces the results in the previous work

Sub issue 1: How does the lambda system of the ESLAC-induced NV-EIT differ from the strain-induced NV-EIT?

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Last year

Calculated degree of mixture of spin

Eigen-basis

$$\{|\psi_{es}\rangle\} = (|A_2\rangle, |A_1\rangle, |E_x\rangle, |E_y\rangle, |E_2\rangle, |E_1\rangle).$$

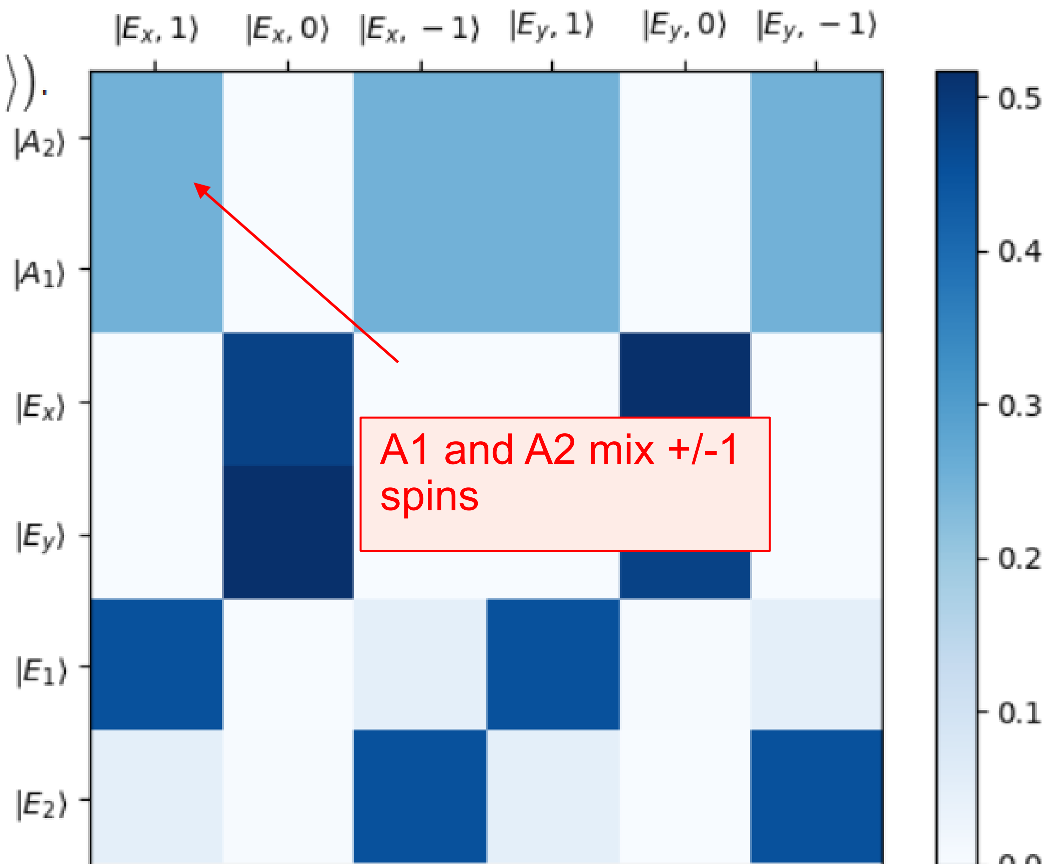
Spin-basis

$$\{|\sigma_{es}\rangle\} = (|E_x, +1\rangle, |E_x, 0\rangle, |E_x, -1\rangle \\ , |E_y, +1\rangle, |E_y, 0\rangle, |E_y, -1\rangle)$$

Mixture

$$|\langle\sigma_{es}|\psi_{es}\rangle|^2$$

B-field 0 G, Strain 0 GHz



Sub issue 1: How does the lambda system of the ESLAC-induced NV-EIT differ from the strain-induced NV-EIT?

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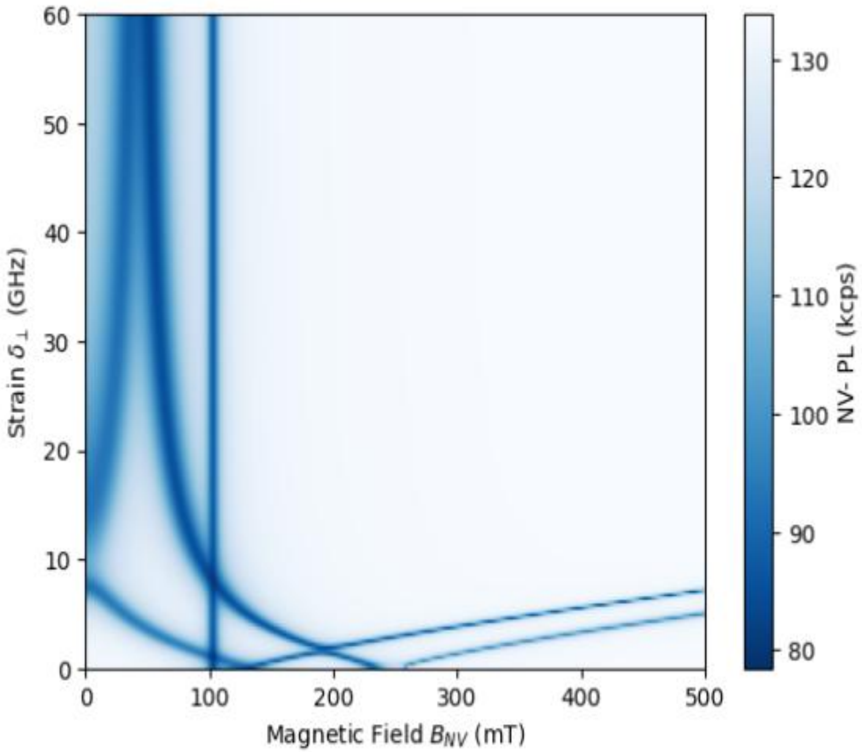
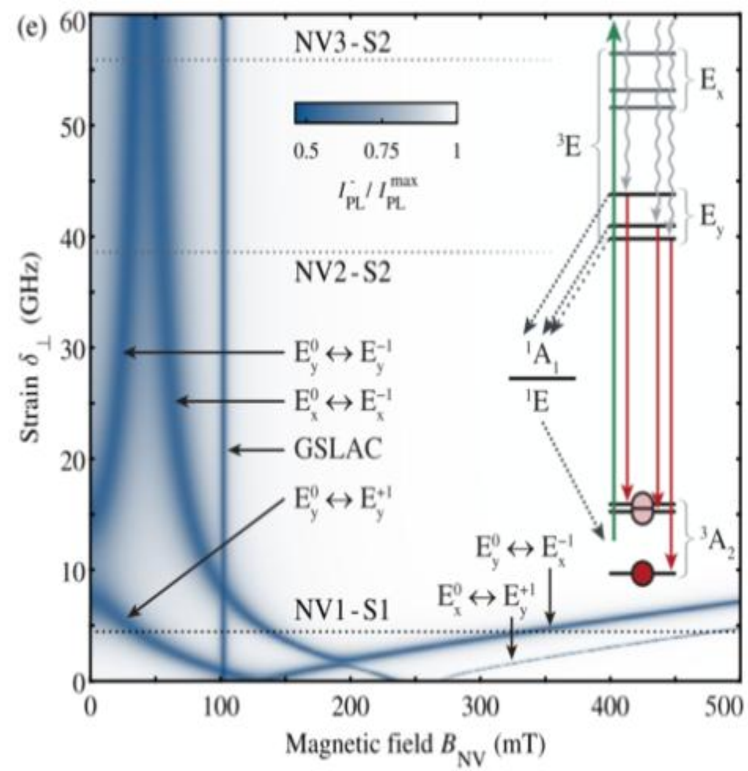
Last year

Calculated ESLAC by Schrodinger equation

Our program



Happacher et al PRL (2022)



Our program reproduces the results in the previous work

How to calculate the degree of spin-basis

$$\{|\psi_{es}\rangle\} = (|A_2\rangle, \underline{|A_1\rangle}, |E_x\rangle, |E_y\rangle, |E_2\rangle, |E_1\rangle).$$

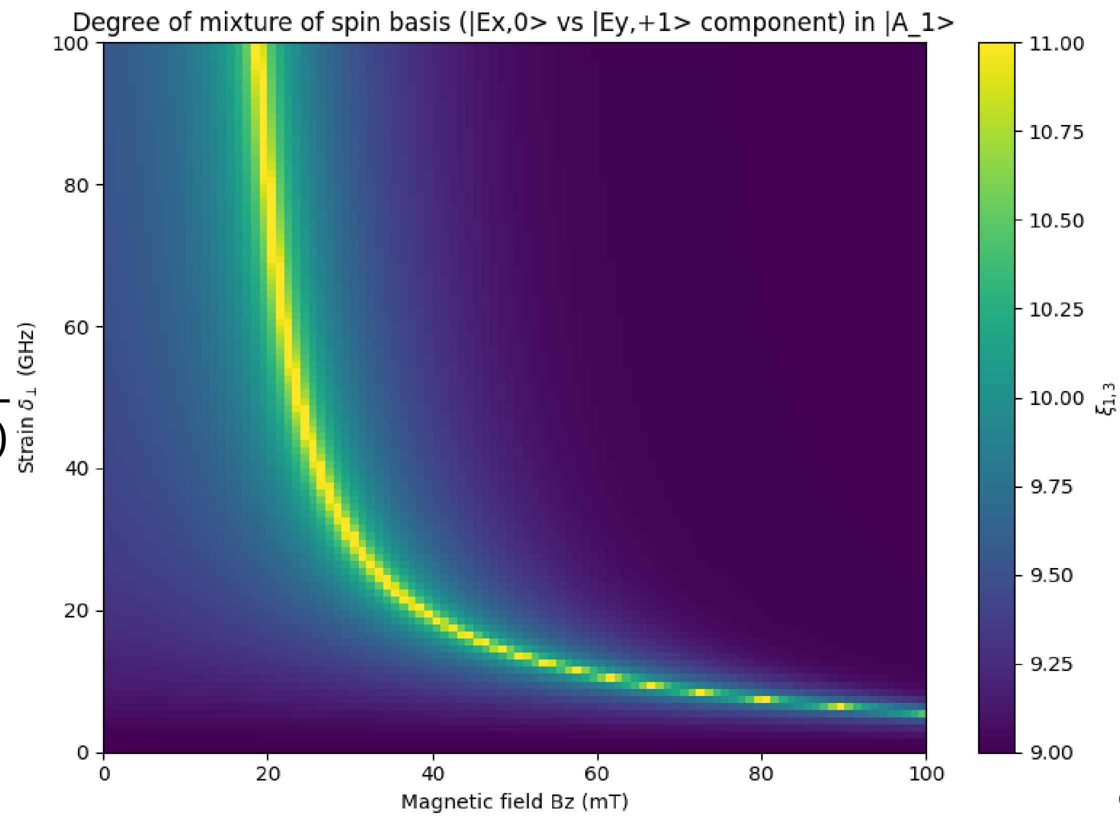
$$\{|\sigma_{es}\rangle\} = (|E_x, +1\rangle, |E_x, 0\rangle, |E_x, -1\rangle \quad |E_y, +1\rangle, |E_y, 0\rangle, |E_y, -1\rangle)$$

Decomposing eigenstates by Spin-bases

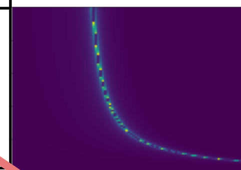
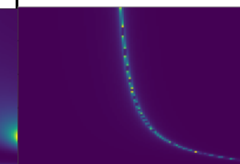
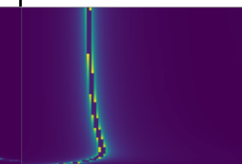
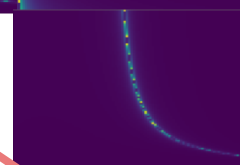
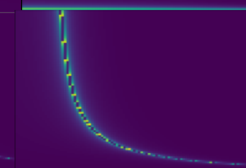
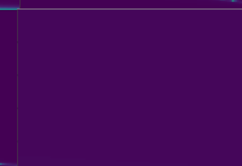
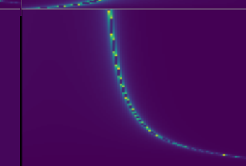
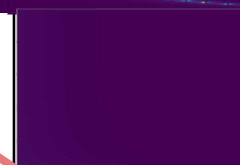
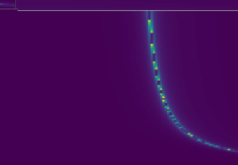
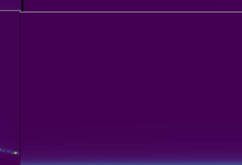
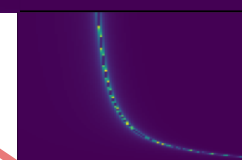
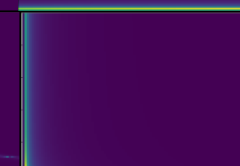
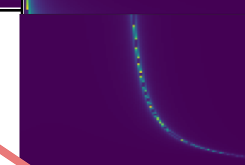
$$|A_1> = a_0|E_x, +1> + a_1|E_x, 0> + a_2|E_x, -1> + a_3|E_y, +1> + a_4|E_y, +1> + a_5|E_y, -1>$$

Definition

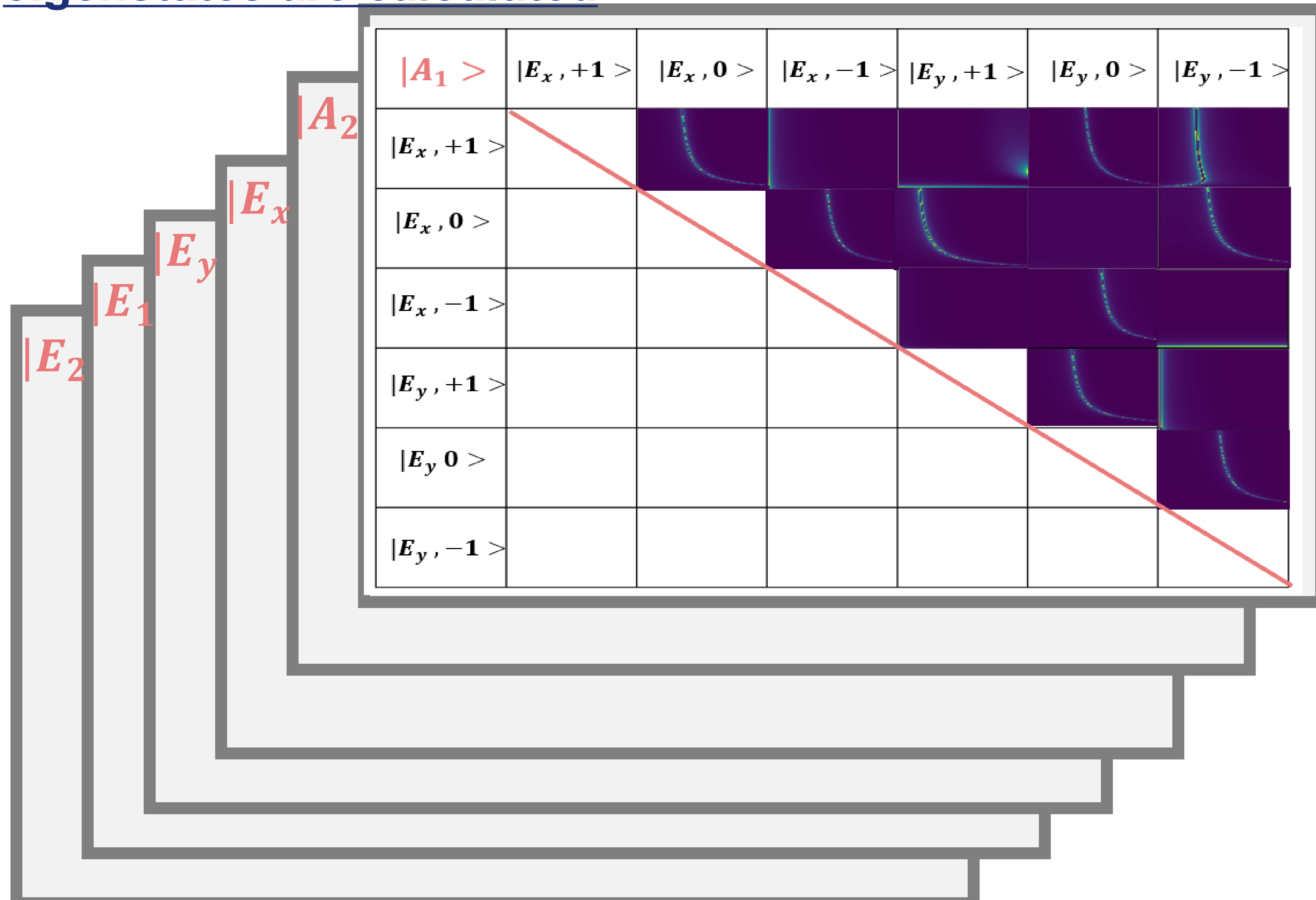
$$\eta(i, j) := In \frac{abs(a_i) + abs(a_j)}{abs(abs(a_i) - abs(a_j))}$$



Mixture of spinbasis in all eigenbases are calculated

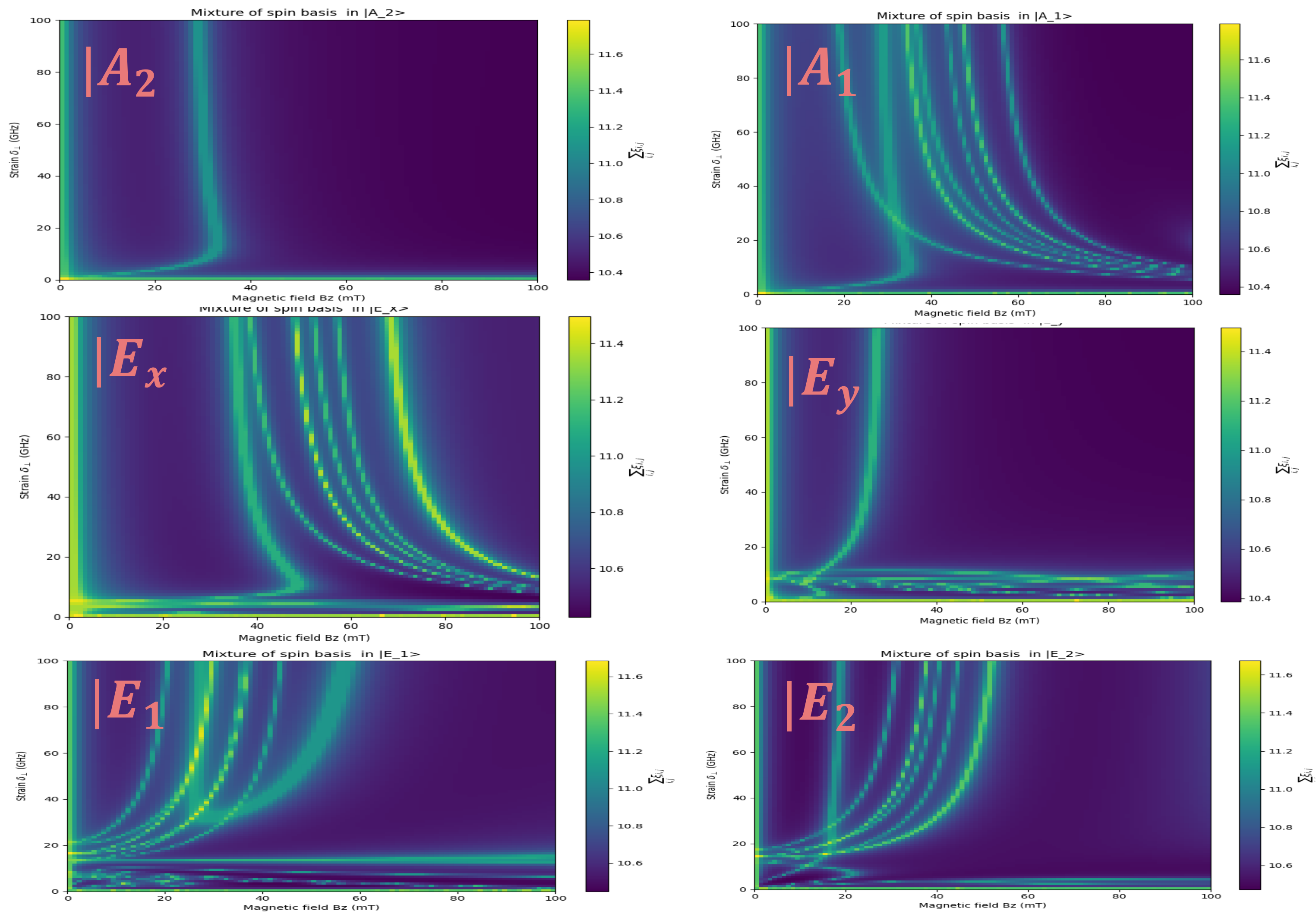
$\vert A_1 >$	$\vert E_x, +1 >$	$\vert E_x, 0 >$	$\vert E_x, -1 >$	$\vert E_y, +1 >$	$\vert E_y, 0 >$	$\vert E_y, -1 >$
$\vert E_x, +1 >$						
$\vert E_x, 0 >$						
$\vert E_x, -1 >$						
$\vert E_y, +1 >$						
$\vert E_y\, 0 >$						
$\vert E_y, -1 >$						

Mixture of spinbasis in all eigenstates are calculated

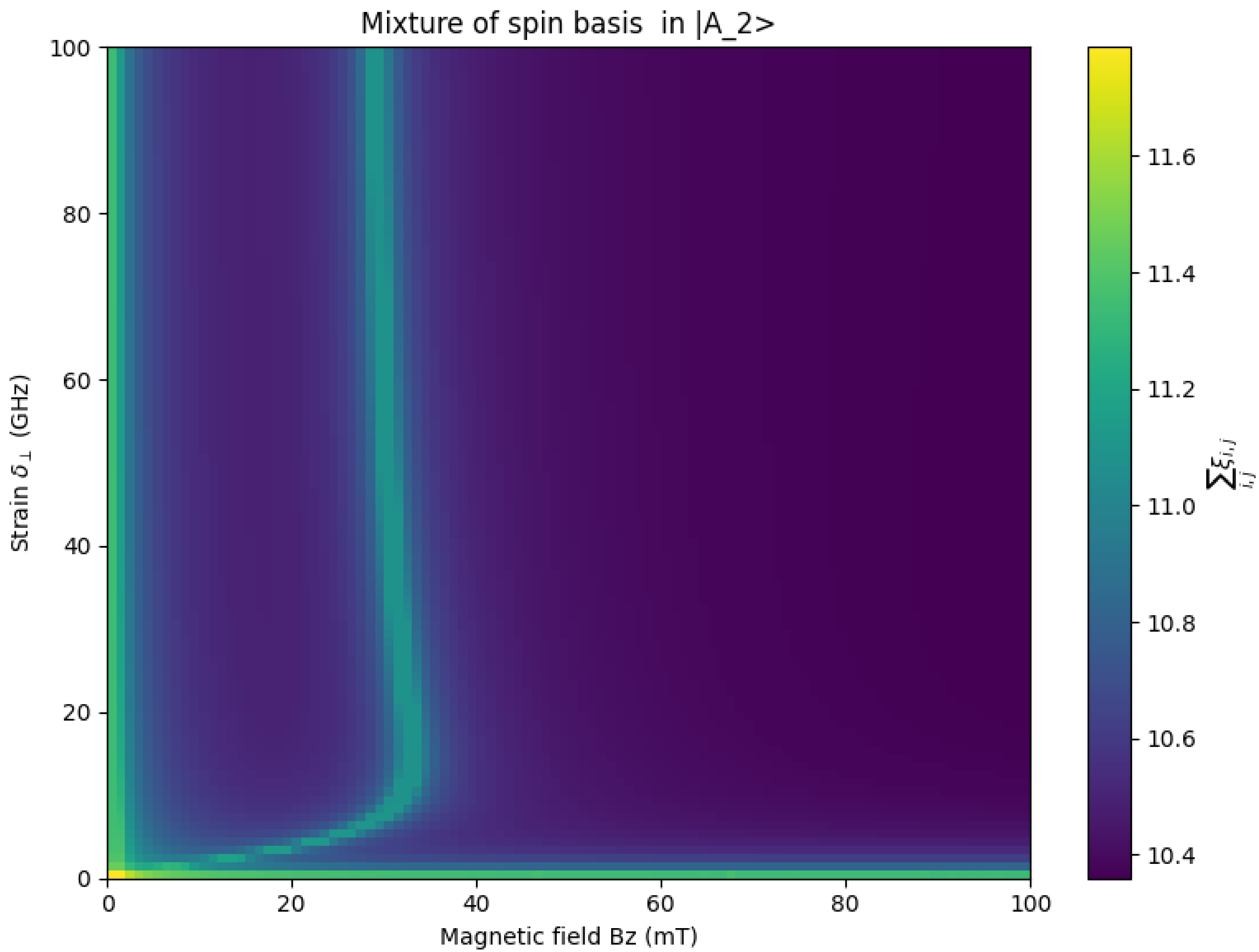


Correct Excited States?

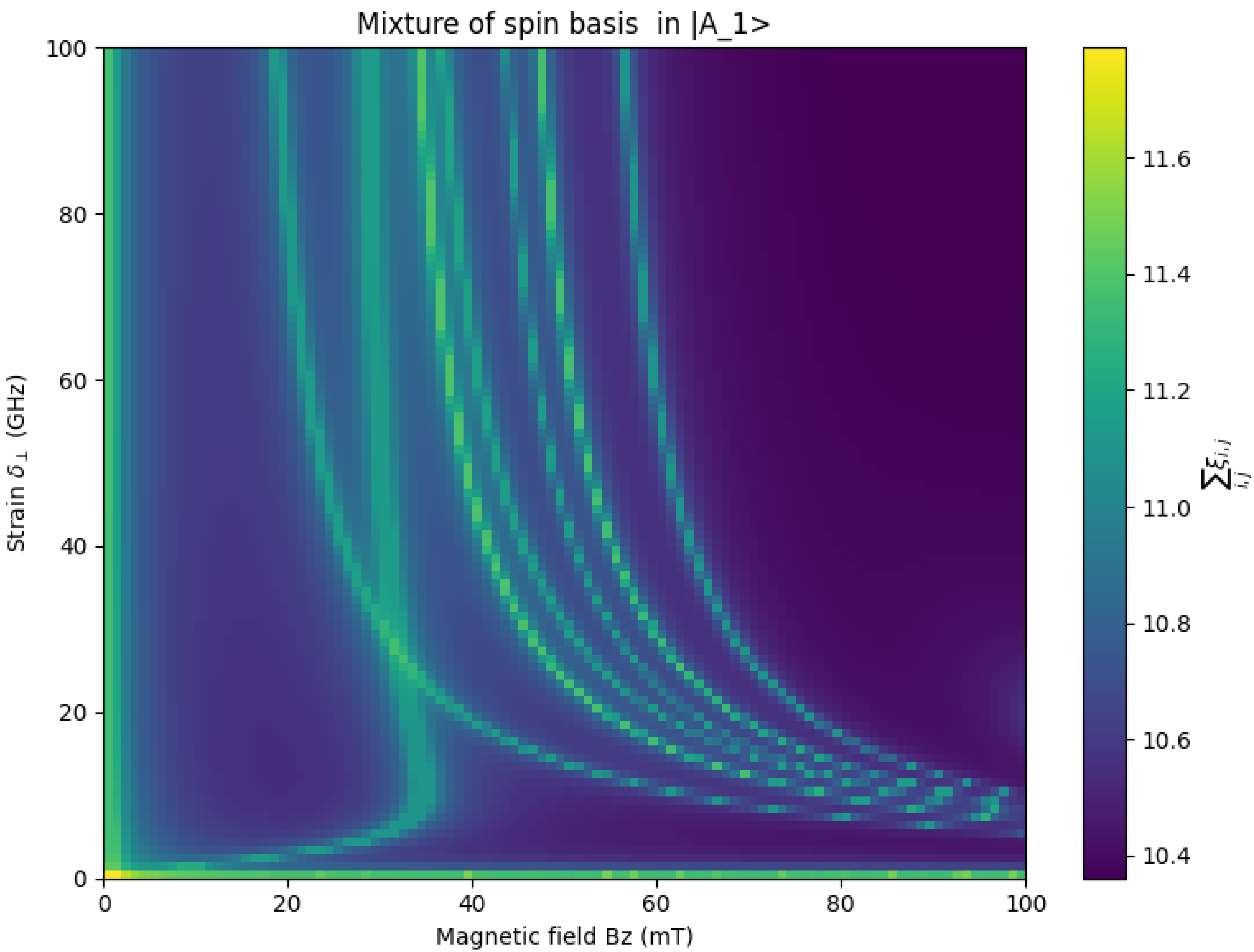
Integrating 15 maps into one



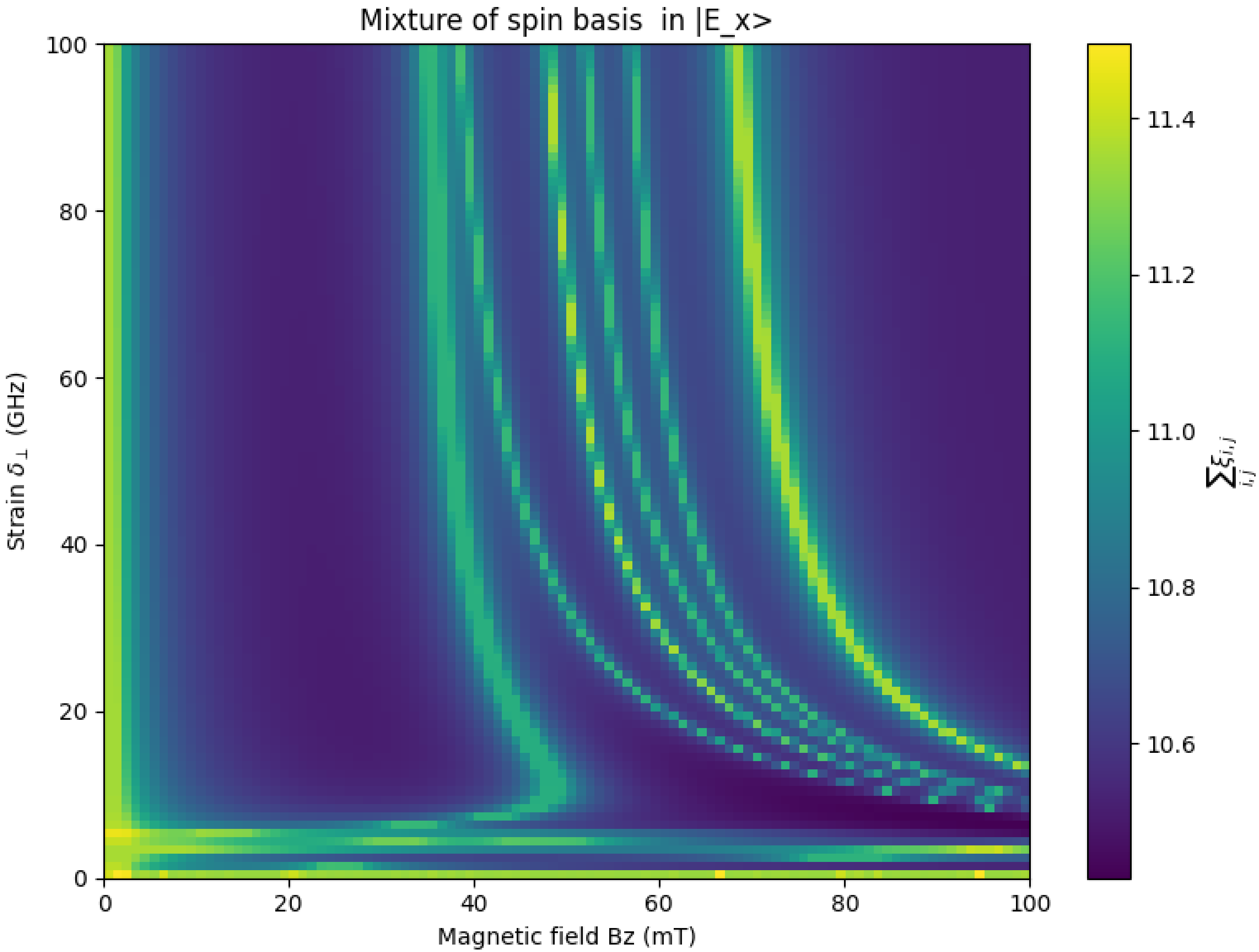
Correct Excited States?



Correct Excited States?



Correct Excited States?



Correct Excited States?

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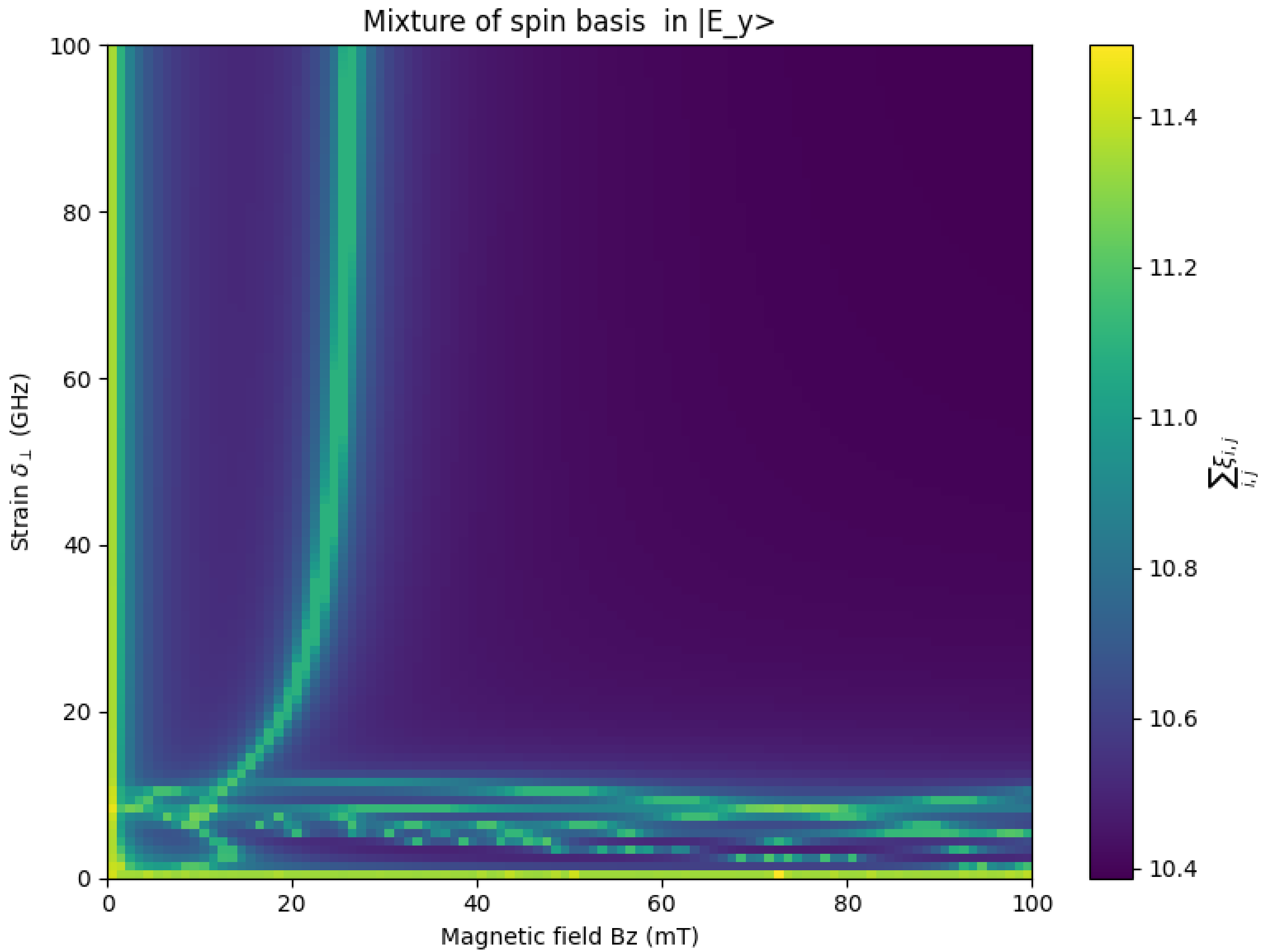
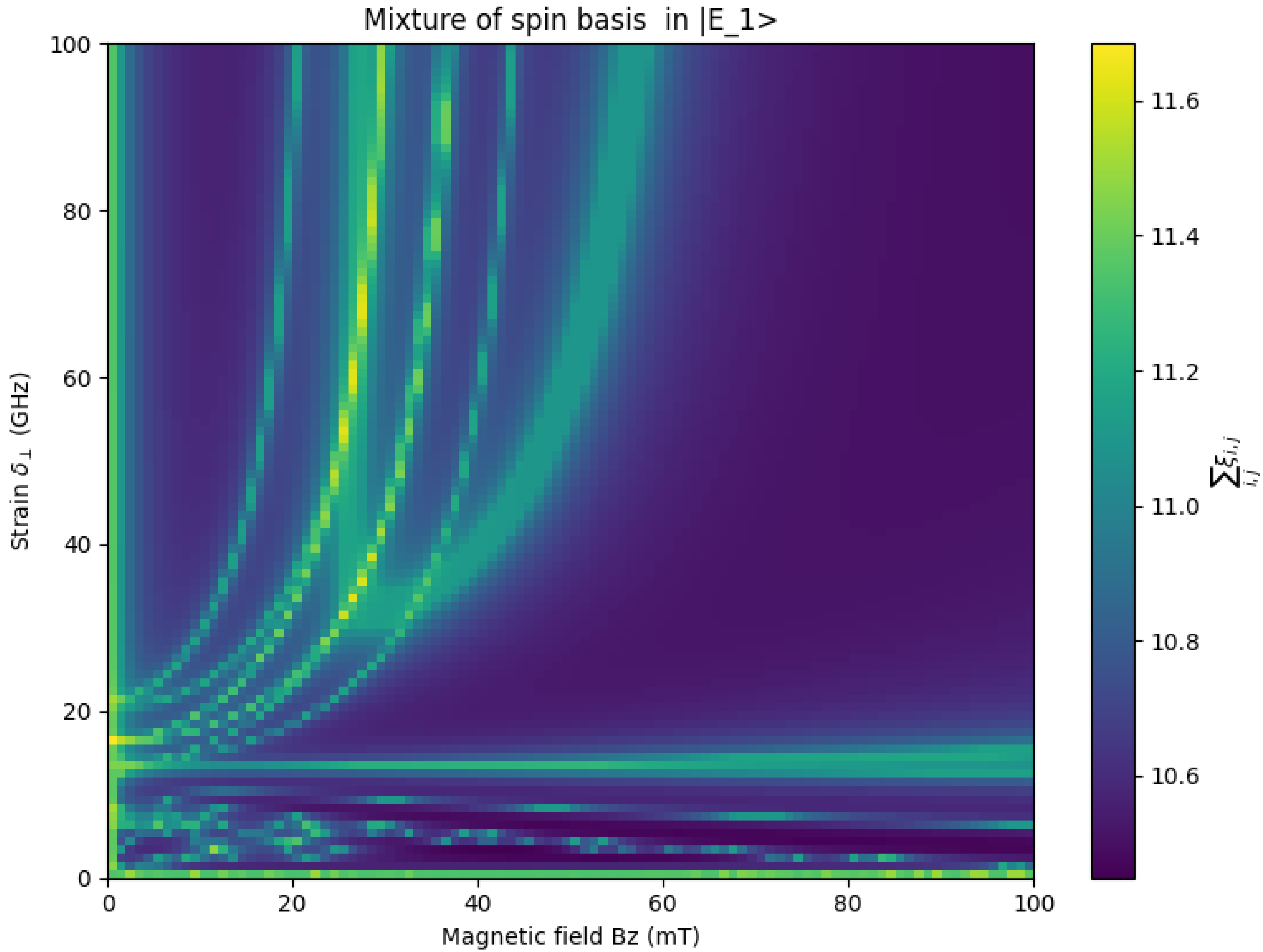
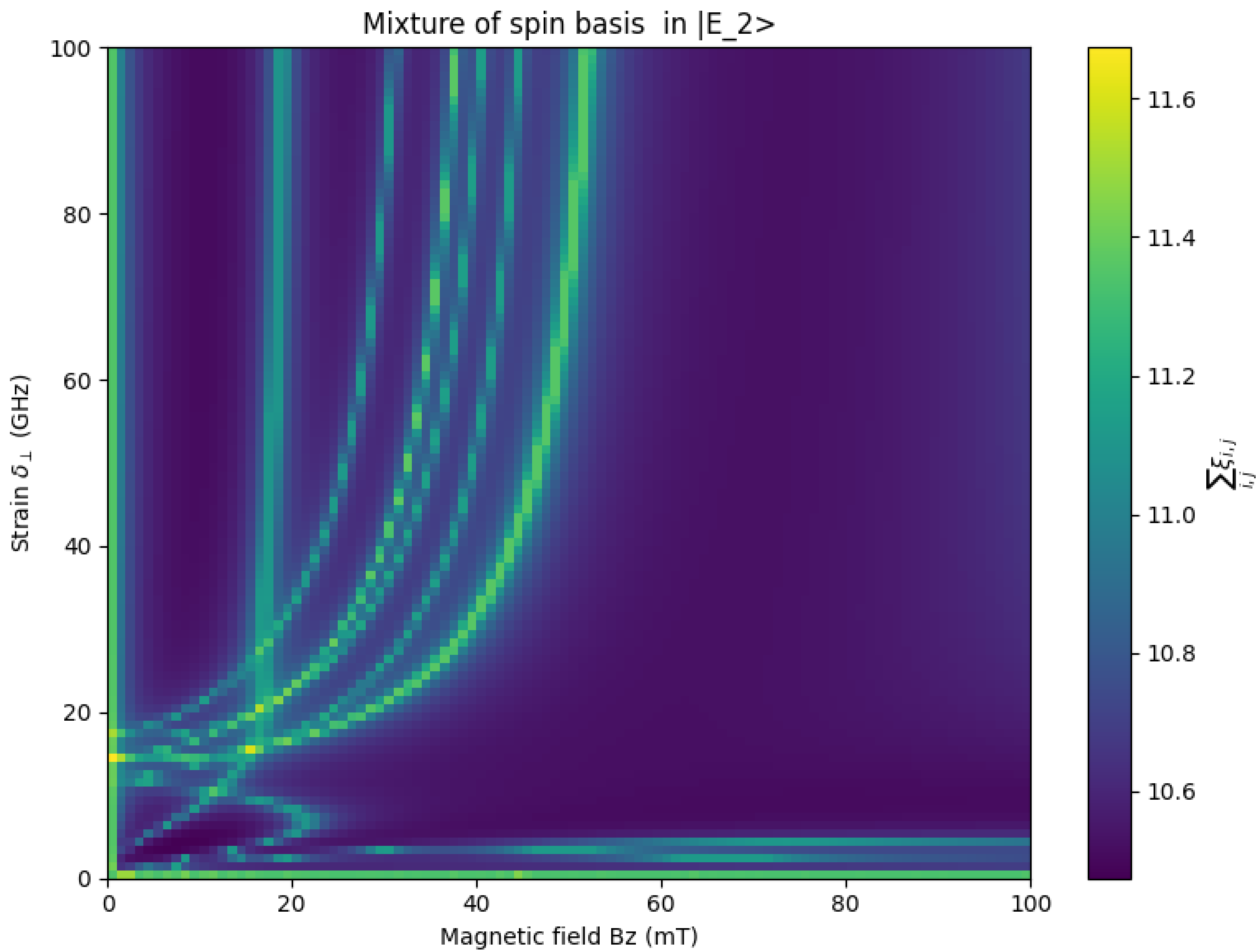


Correct Excited States?



Correct Excited States?

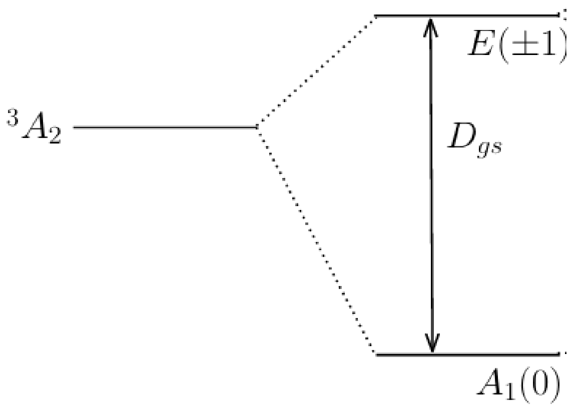


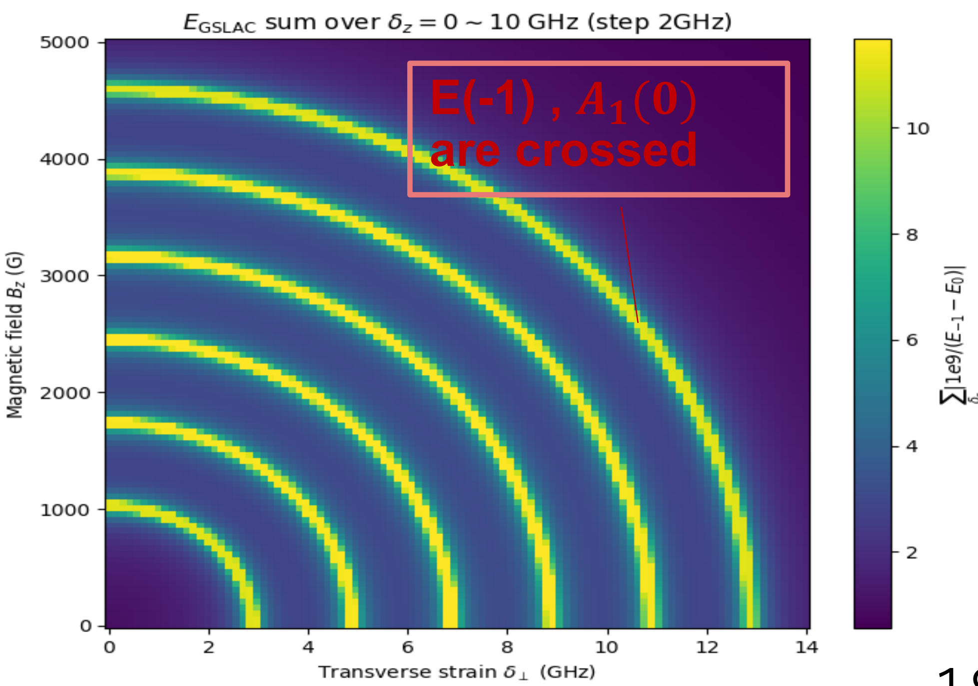
Is GSLAC useable?

From Doherty 2013

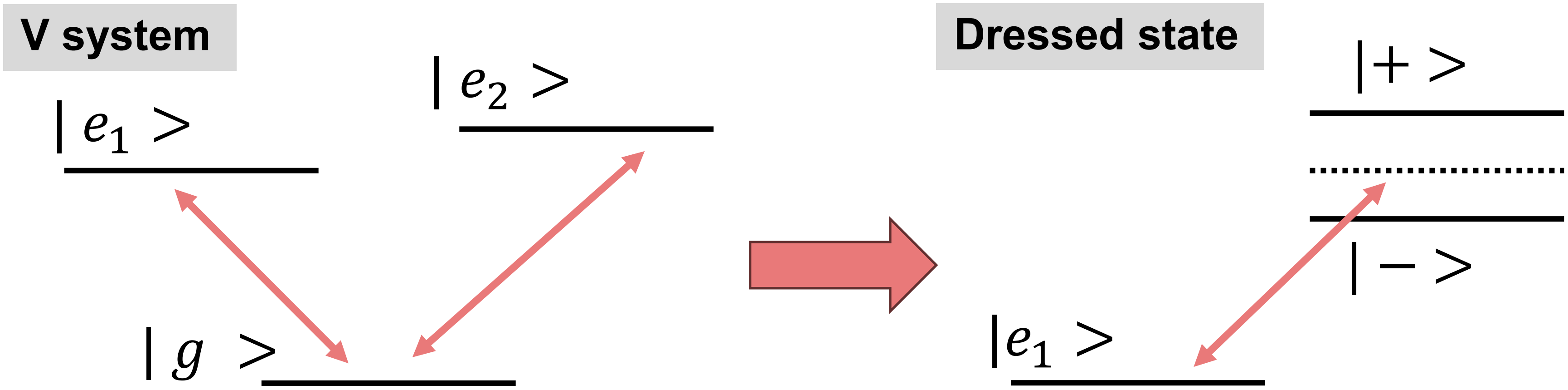
Defined ground-state Hamiltonian at low temperature

$$\begin{aligned}
\hat{H}_{gs} &= D_{gs}\left[\hat{S}_z^2 - S(S+1)/3\right] + A_{gs}^{\parallel}\hat{S}_z\hat{I}_z + A_{gs}^{\perp}\left[\hat{S}_x\hat{I}_x + \hat{S}_y\hat{I}_y\right] \\
&\quad + P_{gs}\left[\hat{I}_z^2 - I(I+1)/3\right] \\
\hat{V}_{gs} &= \mu_B g_{gs}^{\parallel}\hat{S}_z B_z + \mu_B g_{gs}^{\perp}\left(\hat{S}_x B_x + \hat{S}_y B_y\right) + \mu_N g_N \vec{I}\cdot\vec{B} \\
&\quad + d_{gs}^{\parallel}(E_z+\delta_z)\left[\hat{S}_z^2 - S(S+1)/3\right] + d_{gs}^{\perp}(E_x+\delta_x)\left(\hat{S}_y^2 - \hat{S}_x^2\right) \\
&\quad + d_{gs}^{\perp}(E_y+\delta_y)\left(\hat{S}_x\hat{S}_y + \hat{S}_y\hat{S}_x\right)
\end{aligned}$$





Is GSLAC useable?



Key point

・pumping rate < decay rate.(cf. **~83MHz**)

Γ_{pump}~1- 10 MHz   decay rate **~83MHz**

GSLAC is not suitable for EIT

Relevance of the Faraday rotation paper to the room-temperature EIT

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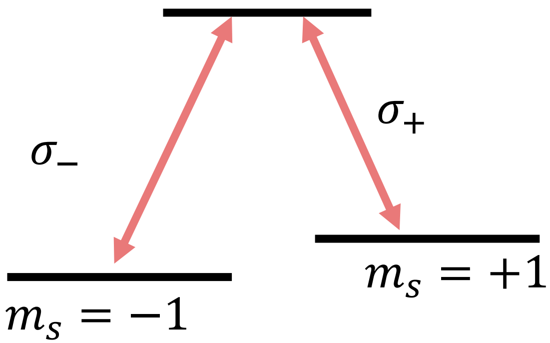
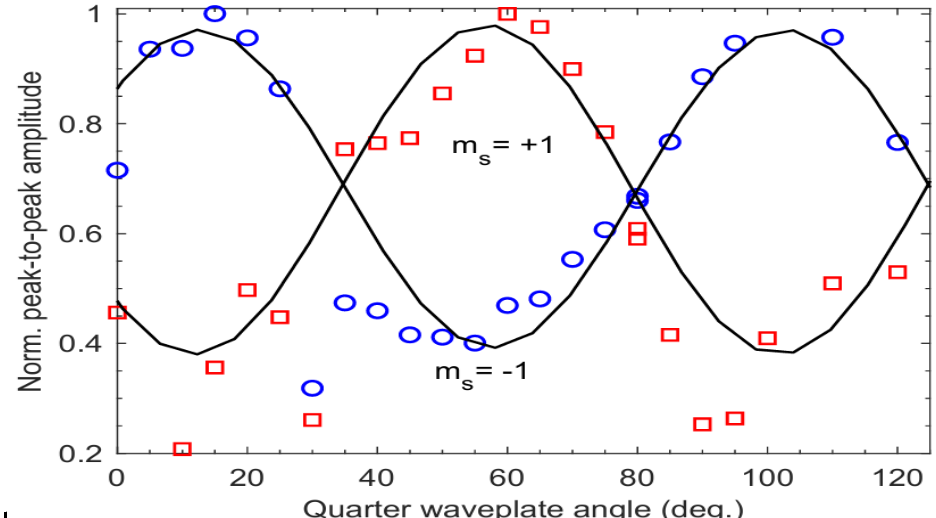
$\{|A_2\rangle, |A_1\rangle\}$



$\sigma_{\pm}$ Right(left)circular polarization light



Reza Kashtibanb 2024

Key points

- It is about confirming three levels systems at room-temperature, via Faraday-effect

Unknown

- How to build this Lambda system
- How can we distinguish between $\{A_1, A_2\}$

Sub issue 2 How robust is the ESLAC-induced EIT against perturbations of B-field and strain?

Sub issue 2: How robust is the ESLAC-induced EIT against perturbations of B-field and strain?

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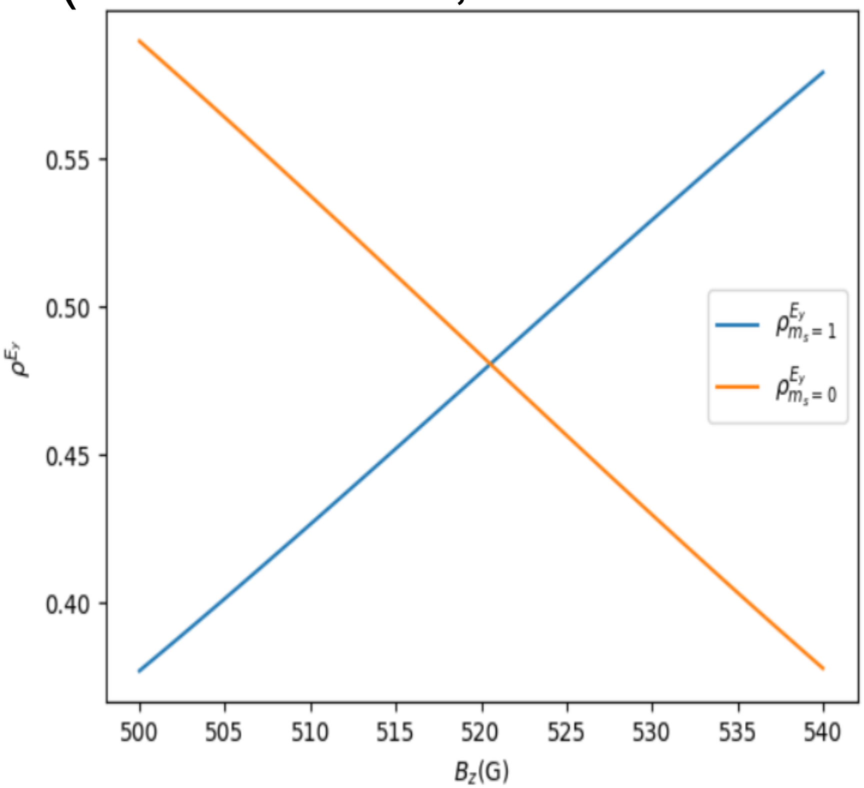
Last year

Calculated change of degree of mixture

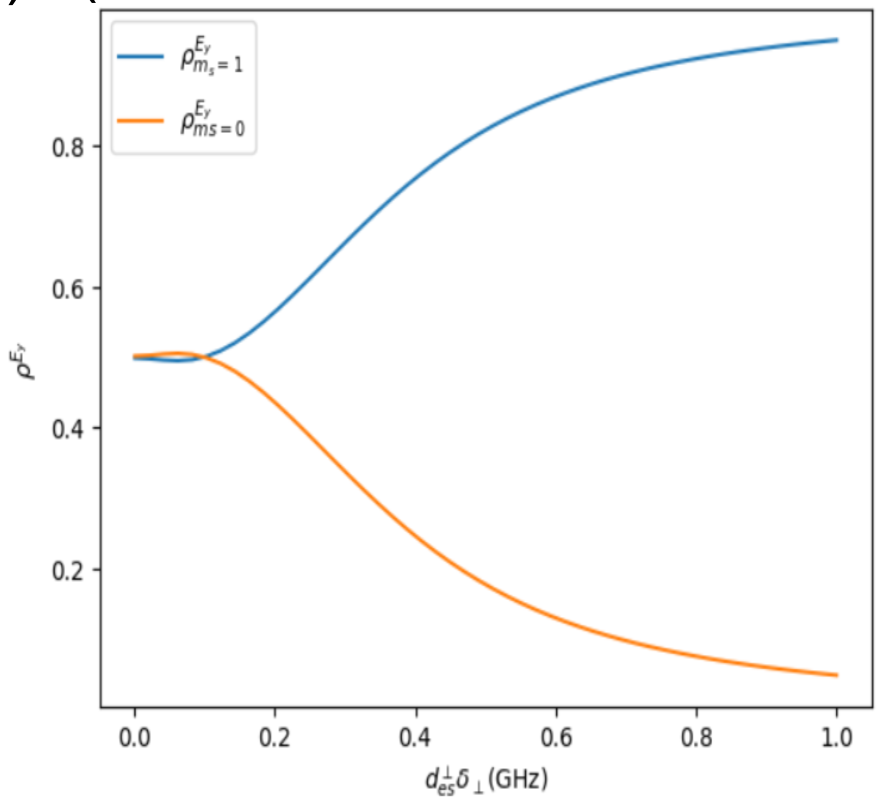
B-field dependence

(B-field 52 mT, Strain 55 GHz)



Strain dependence

(B-field 124 mT, Strain 0 GHz)



Sub issue 2: How robust is the ESLAC-induced EIT against perturbations of B-field and strain?

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Last year

Trying to simulate CPT/EIT by solving Schrodinger equation

Work in progress

This calculation will give us some insights about the EIT signal contrast, which narrows down candidate B-field/strain values.

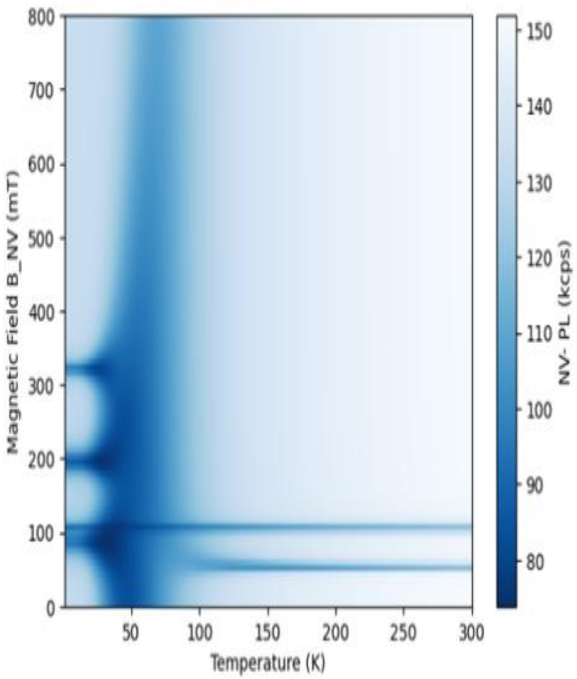
Sub issue 3

Sub issue 3: At which temperature does the ESLAC induced NV-EIT break down?

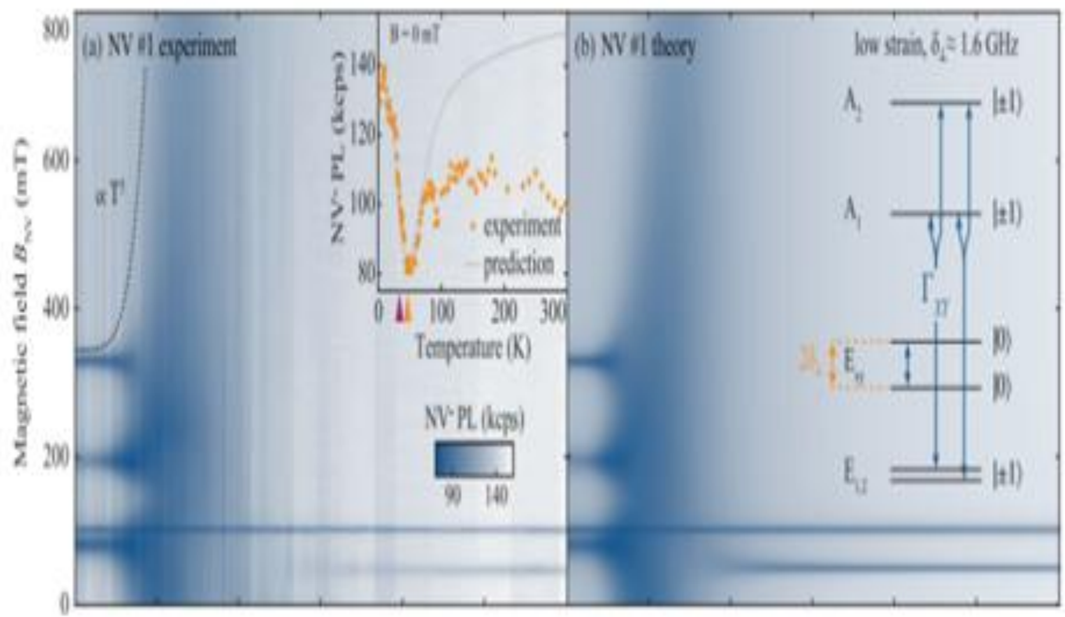
Last year

Calculated photoluminescence at various temperatures

Our program



Happacher et al PRL (2023)



Our program reproduces the results in the previous work

Sub issue 3: At which temperature does the ESLAC induced NV-EIT break down?

Last year

Work in progress

This calculation will tell us up to which temperature NV-EIT signal survives.